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look at the RF performance of the module carefully if an application needs a longer range of, say, 30 metres to 40 metres, in indoor environment or, say, over 200 metres, in outdoor line-of-sight environment with good data throughput."

Kulkarni informs, "Wi-Fi is restricted to 100 metres, while an RF can provide a range up to 700 metres line-of-sight. With a power amplifier, the RF range can be further extended to six kilometres."

Certification. Another angle to look at is certification. Certified modules reduce a fair amount of effort, cost and development time for the designer. Venuturumilli feels, having certified wireless modules in various countries not only enables their customers to sell their end products in various geographies but also provides a time-to-market advantage while reducing risk in the design.

With or without a microcontroller. The user also needs to decide on the microcontroller that needs to be used. Sogani notes, "If a designer opts for a large microcontroller, it is better to go for a module without an embedded microcontroller to make the product cost-effective. If a designer opts for a small microcontroller, it is apt to go for a module that runs bulk of the software such that there is minimal overhead on the microcontroller."

He adds, "For a design without a microcontroller, it would be advisable to go for a module with an embedded microcontroller." Depending on the type of microcontroller and operating system, the decision on the module needs to be made.

Pricing. The word module is being used by various companies for different levels of RF integration, notes Venuturumilli. He says, "Some modules are merely systems in package (SiP), while other modules have all RF components and antennae onboard, and are regulatory-certified." "The challenge customers have with pricing is to make sure they are comparing modules with similar features," he adds.

Power consumption. The amount of power consumed, which significantly depends on the range and purity of data exchange, has been a major focus for hardware designers to incorporate in their designs at the root-level, informs Jha. He says, "Power consumption is a different ball game altogether. Much of the improvements in this area are centred around developing hardware with greater range and higher frequencies and minimising data loss, without any increase in power consumption."

He adds, "Much like in other fields, the wireless community is waiting with bated breath for major breakthroughs in battery technology to help their cause."

Kulkarni feels power consumption of the module is especially important when you need battery-operated modules.

Miscellaneous. Mesh networks, redundancy and data transmission are some other important parameters to consider. "If you need high bandwidth data for transmitting audio/video data, Wi-Fi is apt," says Ganapathy.

Talking about mesh networks, Kulkarni says, "A mesh network would allow the reach to be in-

creased hundred times that of point-to-point star network topology."

Module form factor, functional capability and electrical characteristics are also necessary to consider. Venuturumilli informs, "A buyer must also check if the module is certified in the countries where customer wants to sell the product."

He adds, "Availability of technical support while developing the application and risk of end-of-life by the supplier are two other important aspects."

For future

With CC3000 (and future versions such as CC3100 and CC3200), Hymel informs that, TI created SmartConfig method that allows users to send their Wi-Fi credentials from their smartphones using a series of specially-constructed packets that only CC3x00 can interpret. He says, "As Wi-Fi chips get smaller with more features and solve the connection process problem, we can expect to see more IoT devices in the wild monitoring the air, our health, etc, as well as controlling physical objects from the Internet."

Talking about Redpine Signals' future products, Sogani says, "We are looking at incorporating wireless standards such as 802.11 ah for a longer range in an indoor environment. We are also looking at MIMO and 802.11ac products for higher throughput. For instance, 802.11ac 3X3 mode can give you 1Gbps or higher application throughputs."

He adds, "We also have solutions coming up where microcontrollers will be integrated inside the modules." ●